Delin Qu

Ph.D. Candidate, Fudan University

delinqu.github.io ◇ delinqu.cs@gmail.com ◇ Google Scholar

+86 182 7400 0661 ◇ 2,248 citations, h-index 17

RESEARCH

My research focuses on world action modeling and physical agent reasoning. Currently, I'm working on agentic system design and emergent capabilities in Physical AI, building robots that reason about the physical world to steer fast actions and adapt in long context.

Research Areas: Physical Reasoning Model, Video Action Model, Physical Agentic System.

EXPERIENCE

NVIDIA (Omnimodal World Models for Physical AI, Cosmos 3 & 3.5) — Santa Clara, USA
Research Intern — Jun. 2026 – Present
Advisor: Ming-Yu Liu

- Build unified action spaces for cross-embodiment robot-policy mid-training in Cosmos 3 & 3.5.
- Develop in-context learning and agentic reasoning capabilities for generalist robot policies that reason about the physical world, steer fast actions, and adapt in context.

X-Square Robot (Event World Action Modeling) — Shenzhen, China
Research Intern — Sep. 2025 – Dec. 2025
Advisor: Xiaofan Li

- Built pre-training, fine-tuning, and alignment pipelines for WALL-WM, an event-centered world-action model.
- Evaluated generalization and deployment robustness on real robots.

AgiBot (Reasoning System 2 and Control System 1 Design) — Shanghai, China
Research Intern — Apr. 2025 – Jun. 2025
Advisor: Guanghui Ren

- Proposed OpenSS2, a hierarchical System-1/System-2 policy using phased flow matching.
- Introduced value-guided test-time selection, improving task success by 24% over π_0 on AgiBot G1.

Shanghai AI Laboratory (Open Unified Embodied Foundation Model) — Shanghai, China
Ph.D. Researcher — Aug. 2022 – Present
Advisor: Xuelong Li

- Lead end-to-end research on generalist embodied foundation models, spanning SpatialVLA, EO-1, and Hume.
- Build multimodal data, distributed training, and sim-to-real evaluation pipelines; collaborated with Hugging Face to integrate EO-1 into the official LeRobot codebase.

EDUCATION

Fudan University — Shanghai, China
Ph.D. in Computer Science and Technology — Sep. 2022 – Jun. 2027

Advisor: Prof. Xuelong Li, joint training at Shanghai AI Laboratory

Hunan University — Changsha, China
B.Eng. in Computer Science and Technology — Sep. 2018 – Jun. 2022

GPA: 3.86/4.00; rank: 1/208

SELECTED PUBLICATIONS

- **Cosmos 3: Omnimodal World Models for Physical AI**
  NVIDIA
  Technical Report 2026
  NVIDIA Cosmos World Foundation Model Platform
- **WALL-WM: Carving World Action Modeling at the Event Joints**
  Shalfun Li*, *et al.*, **Delin Qu***, … Qian Wang
  Technical Report 2026 Core contributor
  X-Square Robot World Action Model Technical Report

› **EO-1: Interleaved Vision-Text-Action Pretraining for General Robot Control**
Delin Qu[*], Haoming Song[*], Qizhi Chen, *et al.*, Bin Zhao, Dong Wang, Xuelong Li
IEEE TPAMI 2026
IEEE Transactions on Pattern Analysis and Machine Intelligence

› **SpatialVLA: Exploring Spatial Representations for Vision-Language-Action Models**
Delin Qu[*], Haoming Song[*], Qizhi Chen, *et al.*, Dong Wang, Xuelong Li
RSS 2025 (Spotlight) 658 citations
Robotics: Science and Systems

› **PiX: Pixel Space World Action Model**
Delin Qu[*], Jiaru Li, Bin Zhao, Dong Wang, Xuelong Li
IEEE TPAMI (Under Review)
IEEE Transactions on Pattern Analysis and Machine Intelligence

› **Implicit Event-RGBD Neural SLAM**
Delin Qu[*], Chi Yan[*], Dong Wang, *et al.*, Bin Zhao, Xuelong Li
CVPR 2024 (Spotlight) Top 2.6%
IEEE/CVF Conference on Computer Vision and Pattern Recognition

› **LiveScene: Language Embedding Interactive Radiance Fields for Physical Scene Rendering and Control**
Delin Qu[*], Qizhi Chen[*], *et al.*, Zhigang Wang, Dong Wang, Xuelong Li
NeurIPS 2024
Conference on Neural Information Processing Systems

› **Fast Rolling Shutter Correction in the Wild**
Delin Qu[*], Bangyan Liao[*], Huiqing Zhang, Omar Ait-Aider, Yizhen Lao
IEEE TPAMI 2023
IEEE Transactions on Pattern Analysis and Machine Intelligence

› **Towards Nonlinear-Motion-Aware and Occlusion-Robust Rolling Shutter Correction**
Delin Qu[*], Yizhen Lao, *et al.*, Dong Wang, Bin Zhao, Xuelong Li
ICCV 2023
IEEE/CVF International Conference on Computer Vision

› **Revisiting Rolling Shutter Bundle Adjustment: Toward Accurate and Fast Solution**
Bangyan Liao[*], **Delin Qu**[*], Yifei Xue, Huiqing Zhang, Yizhen Lao
CVPR 2023
IEEE/CVF Conference on Computer Vision and Pattern Recognition

› **Hume: Introducing System-2 Thinking in Vision-Language-Action Models**
Haoming Song[*], **Delin Qu**[*†], *et al.*, Bin Zhao, Dong Wang, Xuelong Li
CVPR 2026 Project Lead
IEEE/CVF Conference on Computer Vision and Pattern Recognition

› **FreeGaussian: Guidance-Free Controllable 3D Gaussian Splats with Flow Derivatives**
Qizhi Chen[*], **Delin Qu**[*†], *et al.*, Dong Wang, Bin Zhao, Xuelong Li
AAAI 2026 (Oral) Top 4.6%; Project Lead
AAAI Conference on Artificial Intelligence

› **GS-SLAM: Dense Visual SLAM with 3D Gaussian Splatting**
Chi Yan[*], **Delin Qu**[*†], *et al.*, Zhigang Wang, Dong Wang, Xuelong Li
CVPR 2024 (Spotlight) Top 2.6%; 776 citations; Project Lead
IEEE/CVF Conference on Computer Vision and Pattern Recognition

SELECTED PROJECTS

OpenPI Comet: Competition Solution for the 2025 BEHAVIOR Challenge *Core Contributor*
Delin Qu[*], Qizhi Chen[*], Shangkun Sun[*], Zhaoshuo Li, *et al.*

Led development of Team Comet's end-to-end $\pi_{0.5}$ training system across multi-dataset pre-training, rejection-sampling post-training, data generation, and evaluation; placed 2nd worldwide in the 2025 BEHAVIOR Challenge, only 0.0085 Q-score behind the winning team.

FastUMI: A Scalable and Hardware-Independent Universal Manipulation Interface with Dataset *Core Contributor*
Zhaxizhuoma, Kehui Liu, *et al.*, **Delin Qu**, Dong Wang, Yan Ding, Bin Zhao, Xuelong Li

Co-developed a scalable, hardware-independent Universal Manipulation Interface for rapid deployment, robust real-world robot data acquisition, and reusable processing workflows.

Any4LeRobot: A Tool Collection for LeRobot *Project Lead, 1.5K stars*
Qizhi Chen, **Delin Qu**, Dong Wang, Bin Zhao

Developed data-conversion, preprocessing, version-migration, and multi-dataset training utilities spanning Open X-Embodiment, AgiBot, RoboMIND, RoboCasa, LIBERO, and RLDS for the LeRobot ecosystem.

BOOK

Bin Zhao, Dong Wang, **Delin Qu**, Haoming Song, Xuelong Li. *Embodied Intelligence: Technology Frontiers and Applications.* Chinese-language book manuscript; under publisher review.

FUNDING & AWARDS

National Natural Science Foundation of China (NSFC) 2024

- Learning to Grow: Interactive Developmental Morphology for Embodied Manipulation Policy
- Motivation: What growth-oriented learning and multi-source interaction mechanisms enable embodied manipulation policies to continually improve in real-world environments?
- Methodology: Develop a spatial-prior-guided action model that improves through a closed loop of planning, execution, feedback, and self-exploration.

2025 BEHAVIOR Challenge 2nd Place Worldwide 2025

Fudan Top Outstanding Ph.D. Student Scholarship Top 1% 2025

Tencent Scholarship Top 0.1% 2023

National Scholarship Top 0.1% 2020, 2021

INVITED TALKS

Real-Time Interactive World Action Models Jun. 2026
Shanghai AI Laboratory, Shanghai

Exploring Spatial Representations for Vision-Language-Action Models Mar. 2025
Institute of Artificial Intelligence, TeleAI / China Telecom

TEACHING & PROFESSIONAL SERVICE

Teaching Assistant, Fudan University 2022 – 2026
COMP130135: Object-Oriented Programming; COMP620002: Artificial Intelligence.

Peer Review, CVPR 2026 Outstanding Reviewer
Reviewer for CVPR, ICCV, ECCV, NeurIPS, ICML, ICLR, TPAMI, and AAAI; AAAI Program Committee member.

TECHNICAL SKILLS

Physical AI Models: VLA models, world models, multimodal pre-training and post-training, in-context learning.

Robot Learning: Generalist robot policies, imitation learning, reinforcement learning, and cross-embodiment transfer.

Learning Systems: Python, PyTorch, Hugging Face Transformers, distributed training, data curation, and evaluation.

3D Perception: Gaussian Splatting, neural rendering, visual SLAM, and real-robot evaluation.

FULL PUBLICATION LIST

2026

- OpenFlyV2: An onboard vision-language-action model for aerial perception and navigation
  Junli Liu, *et al.*, **Delin Qu**, ... Xuelong Li
  arXiv 2026 (submitted)
  arXiv preprint
- Cosmos 3: Omnimodal World Models for Physical AI
  NVIDIA
  Technical Report 2026
  NVIDIA Cosmos World Foundation Model Platform
- WALL-WM: Carving World Action Modeling at the Event Joints
  Shalfun Li*, Victor Yao*, Charles Yang*, **Delin Qu***, *et al.*, Qian Wang
  arXiv 2026
  arXiv preprint
- EO-1: Interleaved Vision-Text-Action Pretraining for General Robot Control
  Delin Qu*, Haoming Song*, Qizhi Chen, *et al.*, Bin Zhao, Dong Wang, Xuelong Li
  IEEE TPAMI 2026
  IEEE Transactions on Pattern Analysis and Machine Intelligence

› FM-Steer: Enhance Generalist Policies with Value-Guided Cascaded Denoising
Haoming Song*, **Delin Qu***, *et al.*, Bin Zhao, Dong Wang, Xuelong Li
CVPR 2026 (formerly Hume)
IEEE/CVF Conference on Computer Vision and Pattern Recognition

› FreeGaussian: Annotation-Free Control of Articulated Objects via 3D Gaussian Splats with Flow Derivatives
Qizhi Chen*, **Delin Qu***†, Junli Liu, *et al.*, Yuan Yuan, Bin Zhao
AAAI 2026 (Oral)
AAAI Conference on Artificial Intelligence

› Exploring the Potential of Encoder-Free Architectures in 3D LMMs
Yiwen Tang*, Zoey Guo*, Zhuhao Wang*, Ray Zhang*, Qizhi Chen, Junli Liu, **Delin Qu**, *et al.*
ICLR 2026
International Conference on Learning Representations

› FVAR: Next-Focus Prediction for Visual Autoregressive Modeling
Xiaofan Li, Chenming Wu, Yanpeng Sun, Jiaming Zhou, **Delin Qu**, *et al.*
CVPR 2026
IEEE/CVF Conference on Computer Vision and Pattern Recognition

› Are We Ready for RL in Text-to-3D Generation? A Progressive Investigation
Yiwen Tang, Zoey Guo, Kaixin Zhu, *et al.*, **Delin Qu**
CVPR 2026
IEEE/CVF Conference on Computer Vision and Pattern Recognition

› Q-GeoMem: Question-Guided Geometric Memory for Video Spatial Reasoning
Xianqiang Gao, Qizhi Chen, **Delin Qu**, *et al.*, Xuelong Li
arXiv 2026
arXiv preprint

› VGGT-Edit: Feed-Forward Native 3D Scene Editing with Residual Field Prediction
Kaixin Zhu, Yiwen Tang, Yifan Yang, *et al.*, **Delin Qu**
arXiv 2026
arXiv preprint

› PiX: Pixel Space World Action Model
Delin Qu*, Jiaru Li, Bin Zhao, Dong Wang, Xuelong Li
IEEE TPAMI (Under Review)
IEEE Transactions on Pattern Analysis and Machine Intelligence

2025

› SpatialVLA: Exploring Spatial Representations for Vision-Language-Action Models
Delin Qu*, Haoming Song*, Qizhi Chen, *et al.*, Dong Wang, Xuelong Li
RSS 2025 (Spotlight)
Robotics: Science and Systems

› Learning 2D Invariant Affordance Knowledge for 3D Affordance Grounding
Xianqiang Gao, Pingrui Zhang, **Delin Qu**, *et al.*, Bin Zhao
AAAI 2025 (Oral)
AAAI Conference on Artificial Intelligence

› Think Small, Act Big: Primitive Prompt Learning for Lifelong Robot Manipulation
Yuanqi Yao, Siao Liu, Haoming Song, **Delin Qu**, *et al.*, Dong Wang
CVPR 2025
IEEE/CVF Conference on Computer Vision and Pattern Recognition

› Revisiting Multi-Agent World Modeling from a Diffusion-Inspired Perspective
Yang Zhang, Xinran Li, Jianing Ye, Shuang Qiu, **Delin Qu**, *et al.*
NeurIPS 2025
Conference on Neural Information Processing Systems

› FastUMI: A Scalable and Hardware-Independent Universal Manipulation Interface with Dataset
Zhaxizhuoma, Kehui Liu, Chuyue Guan, *et al.*, **Delin Qu**
CoRL 2025
Conference on Robot Learning

› F1: A Vision-Language-Action Model Bridging Understanding and Generation to Actions
Qi Lv, Weijie Kong, Hao Li, *et al.*, **Delin Qu**
arXiv 2025
arXiv preprint

› UniF^{2}ace: A Unified Fine-Grained Face Understanding and Generation Model
Junzhe Li, Sifan Zhou, Liya Guo, *et al.*, **Delin Qu**
arXiv 2025
arXiv preprint

› Alias-Free 4D Gaussian Splatting
Zilong Chen, Huan-ang Gao, **Delin Qu**, *et al.*, Hao Zhao
arXiv 2025
arXiv preprint

› OpenPI Comet: Competition Solution for the 2025 BEHAVIOR Challenge
Junjie Bai, Yu-Wei Chao, Qizhi Chen, *et al.*, **Delin Qu**
arXiv 2025
arXiv preprint

› Ego-Motion Estimation for Vehicles with a Rolling Shutter Camera
Yongcong Zhang, Bangyan Liao, **Delin Qu**, *et al.*, Yizhen Lao
IEEE T-IV 2025
IEEE Transactions on Intelligent Vehicles

2024

› LiveScene: Language Embedding Interactive Radiance Fields for Physical Scene Control and Rendering
Delin Qu[*], Qizhi Chen[*], *et al.*, Zhigang Wang, Dong Wang, Xuelong Li
NeurIPS 2024
Conference on Neural Information Processing Systems

› Implicit Event-RGBD Neural SLAM
Delin Qu[*], Chi Yan[*], Dong Wang, *et al.*, Bin Zhao, Xuelong Li
CVPR 2024 (Spotlight)
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› GS-SLAM: Dense Visual SLAM with 3D Gaussian Splatting
Chi Yan[*], **Delin Qu**[*†], *et al.*, Zhigang Wang, Dong Wang, Xuelong Li
CVPR 2024 (Spotlight)
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2023

› Fast Rolling Shutter Correction in the Wild
Delin Qu[*], Bangyan Liao[*], Huiqing Zhang, Omar Ait-Aider, Yizhen Lao
IEEE TPAMI 2023
IEEE Transactions on Pattern Analysis and Machine Intelligence

› Towards Nonlinear-Motion-Aware and Occlusion-Robust Rolling Shutter Correction
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› Revisiting Rolling Shutter Bundle Adjustment: Toward Accurate and Fast Solution
Bangyan Liao[*], **Delin Qu**[*], Yifei Xue, Huiqing Zhang, Yizhen Lao
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